

SkillDzire Advanced VLSI & Chip Design

1. VLSI stands for:

- a) very large signal integration
- b) very large scale integration**
- c) variable logic system integration
- d) vertical logic scale integration

2. Which technology enabled VLSI growth the most?

- a) vacuum tubes
- b) bipolar transistors
- c) mosfet technology**
- d) relays

3. Moore's Law predicts:

- a) power increases exponentially
- b) cost increases with size
- c) transistors double every ~18–24 months**
- d) clock speed remains constant

4. Which material is most commonly used for IC fabrication?

- a) germanium
- b) gallium arsenide
- c) silicon**
- d) copper

5. SSI circuits typically contain:

- a) less than 10 gates**
- b) 10–100 gates
- c) 100–1000 gates
- d) more than 10000 gates

6. Which integration level includes microprocessors?

- a) ssi
- b) msi
- c) lsi
- d) vlsi**

7. CMOS technology is preferred because it offers:

- a) high static power
- b) low noise immunity
- c) low power consumption**
- d) complex fabrication

8. Semiconductor devices work based on:

- a) insulation
- b) conduction and insulation
- c) controlled conductivity**
- d) superconductivity

9. Which node refers to transistor gate length?

- a) technology node**
- b) power node
- c) logic node
- d) clock node

10. FinFET technology mainly reduces:

- a) chip size
- b) leakage current**
- c) clock skew
- d) packaging cost

11. RTL design belongs to:

- a) backend
- b) frontend**
- c) fabrication
- d) packaging

12. Which language is mainly used in frontend design?

- a) c
- b) python
- c) verilog/vhdl**
- d) spice

13. Synthesis converts:

- a) layout to netlist
- b) rtl to gate-level netlist**
- c) netlist to gdsii
- d) gdsii to silicon

14. Place and Route is part of:

- a) frontend
- b) backend**
- c) verification
- d) testing

15. Static Timing Analysis (STA) is done mainly in:

- a) rtl coding
- b) logic synthesis
- c) backend flow**
- d) fabrication

16. GDSII file is generated during:

- a) rtl design
- b) synthesis
- c) physical design**
- d) verification

17. Clock Tree Synthesis (CTS) aims to reduce:

- a) power
- b) delay
- c) clock skew**
- d) area

18. DRC and LVS checks belong to:

- a) frontend verification
- b) backend verification**
- c) functional simulation
- d) rtl verification

19. Logical verification ensures:

- a) layout correctness
- b) electrical rules
- c) functional correctness**
- d) fabrication yield

20. Parasitic extraction is done after:

- a) rtl design
- b) synthesis
- c) placement and routing**
- d) packaging

21. A combinational circuit output depends on:

- a) clock
- b) present inputs only**
- c) previous outputs
- d) memory elements

22. Which is NOT a combinational circuit?

- a) decoder
- b) multiplexer
- c) counter**
- d) encoder

23. AND gate output is HIGH when:

- a) any input is high
- b) all inputs are high**
- c) inputs differ
- d) all inputs are low

24. Truth table represents:

- a) physical layout
- b) voltage levels
- c) all possible input-output combinations**
- d) timing constraints

25. A half adder produces:

- a) **sum and carry**
- b) sum and borrow
- c) difference and carry
- d) product and quotient

26. XOR gate outputs HIGH when:

- a) inputs are same
- b) **inputs are different**
- c) all inputs are high
- d) all inputs are low

27. Number of rows in a truth table for 3 inputs is:

- a) 3
- b) 6
- c) **8**
- d) 9

28. Which gate is called a universal gate?

- a) and
- b) or
- c) **nand**
- d) xor

29. De Morgan's theorem helps in:

- a) power reduction
- b) **logic simplification**
- c) layout design
- d) clock design

30. A multiplexer selects:

- a) **one output from many inputs**
- b) many outputs from one input
- c) clock signals
- d) memory locations

31. K-map is used to:

- a) verify timing
- b) simplify boolean expressions**
- c) draw layouts
- d) generate clocks

32. A 4:1 MUX has how many select lines?

- a) 1
- b) 2**
- c) 3
- d) 4

33. SOP form stands for:

- a) sum of products**
- b) system of products
- c) sequence of processes
- d) sum of powers

34. POS form stands for:

- a) product of sums**
- b) power of signals
- c) process of synthesis
- d) product of systems

35. Boolean expression minimization helps reduce:

- a) delay and area**
- b) voltage
- c) frequency
- d) noise

36. Which circuit converts binary to one-hot output?

- a) encoder
- b) decoder**
- c) comparator
- d) adder

37. A full adder has how many inputs?

- a) 2
- b) 3**
- c) 4
- d) 5

38. Logic optimization improves:

- a) yield only
- b) power, area, and speed**
- c) fabrication cost only
- d) packaging

39. Hazard in combinational circuits causes:

- a) permanent failure
- b) temporary glitches**
- c) power loss
- d) clock skew

40. Which logic has minimum gate count?

- a) unsimplified logic
- b) k-map optimized logic**
- c) random logic
- d) truth table

41. A latch is:

- a) edge-triggered
- b) level-sensitive**
- c) clock-free
- d) memoryless

42. A flip-flop is:

- a) level-sensitive
- b) asynchronous
- c) edge-triggered**
- d) analog

43. Which is NOT a flip-flop type?

- a) sr
- b) jk
- c) d
- d) xor**

44. Setup time refers to:

- a) clock width
- b) data stable before clock edge**
- c) data stable after clock edge
- d) output delay

45. Hold time refers to:

- a) data stable after clock edge**
- b) data stable before clock edge
- c) clock period
- d) reset time

46. Violation of setup time causes:

- a) metastability**
- b) reduced power
- c) faster operation
- d) no effect

47. D flip-flop stores data at:

- a) any time
- b) falling edge
- c) rising edge**
- d) both edges

48. JK flip-flop avoids:

- a) race condition**
- b) toggle operation
- c) clocking
- d) memory

49. Sequential circuits contain:

- a) only logic gates
- b) memory elements**
- c) no clock
- d) only combinational logic

50. Metastability occurs when:

- a) clock fails
- b) setup/hold times are violated**
- c) power is high
- d) logic is minimized

51. Sequential circuits differ from combinational circuits because they:

- a) have no logic gates
- b) depend on present input and past state**
- c) do not use clocks
- d) have no memory

52. The memory element in sequential circuits is:

- a) logic gate
- b) multiplexer
- c) flip-flop**
- d) decoder

53. Clock signal is used to:

- a) reduce power
- b) synchronize state changes**
- c) increase delay
- d) generate logic

54. Which is a synchronous sequential circuit?

- a) latch-based circuit
- b) flip-flop-based circuit**
- c) diode logic
- d) combinational logic

55. State of a sequential circuit is stored in:

- a) inputs
- b) outputs
- c) flip-flops**
- d) combinational logic

56. Feedback is essential in:

- a) combinational circuits
- b) sequential circuits**
- c) decoders
- d) multiplexers

57. Finite State Machines (FSMs) are used to model:

- a) combinational logic
- b) sequential behavior**
- c) analog circuits
- d) power systems

58. The number of states depends on:

- a) number of inputs only
- b) number of flip-flops used**
- c) clock frequency
- d) output logic

59. State diagram represents:

- a) physical layout
- b) truth table
- c) state transitions graphically**
- d) RTL code

60. State table shows:

- a) only outputs
- b) present state, next state, outputs**
- c) only inputs
- d) timing constraints

61. Reset is used to:

- a) increase speed
- b) reduce power
- c) initialize the circuit state**
- d) generate clock

62. Asynchronous reset means:

- a) reset independent of clock**
- b) reset only at clock edge
- c) reset after delay
- d) reset through software

63. Synchronous reset occurs:

- a) anytime
- b) at active clock edge**
- c) before clock
- d) without clock

64. Race condition occurs due to:

- a) slow clock
- b) unequal propagation delays**
- c) low voltage
- d) excess power

65. Proper state encoding helps in:

- a) increasing glitches
- b) reducing logic complexity**
- c) increasing delay
- d) power wastage

66. Mealy machine output depends on:

- a) present state only
- b) present state and input**
- c) next state only
- d) clock only

67. Moore machine output depends on:

- a) present state only**
- b) present input only
- c) input and output
- d) clock

68. Which machine generally has fewer states?

- a) mealy machine**
- b) moore machine
- c) combinational machine
- d) latch

69. Output change in Mealy machine is:

- a) faster**
- b) slower
- c) clock independent
- d) delayed by one cycle

70. Moore machine output changes:

- a) immediately
- b) only on state transition**
- c) asynchronously
- d) without clock

71. Mealy machine outputs are associated with:

- a) transitions**
- b) states
- c) flip-flops
- d) inputs only

72. Moore machine outputs are associated with:

- a) transitions
- b) states**
- c) inputs
- d) clock

73. Glitches are more likely in:

- a) mealy machines**
- b) moore machines
- c) counters
- d) registers

74. FSM with stable outputs is best implemented as:

- a) mealy
- b) moore**
- c) combinational
- d) asynchronous

75. State reduction aims to:

- a) increase power
- b) increase states
- c) minimize number of states**
- d) slow operation

76. FSM is classified based on:

- a) clock frequency
- b) output dependency**
- c) power consumption
- d) layout style

77. FSM modeling is commonly used in:

- a) power supplies
- b) control units**
- c) analog circuits
- d) antennas

78. Mealy machine needs fewer flip-flops because:

- a) it uses fewer states**
- b) it uses more memory
- c) it avoids clock
- d) it is asynchronous

79. Moore machine is preferred when:

- a) speed is critical
- b) output stability is required**
- c) glitches are acceptable
- d) power is unlimited

80. FSM design starts with:

- a) RTL coding
- b) state diagram**
- c) synthesis
- d) layout

81. First step in Mealy machine design is:

- a) write Verilog
- b) define states and transitions**
- c) synthesis
- d) testing

82. Mealy state diagram labels include:

- a) state/output
- b) input/output**
- c) output only
- d) input only

83. Next-state logic is derived from:

- a) state table**
- b) layout
- c) clock
- d) testbench

84. Output logic in Mealy machine depends on:

- a) state and input**
- b) state only
- c) clock only
- d) reset

85. State encoding converts:

- a) outputs to states
- b) states to binary values**
- c) inputs to outputs
- d) logic to layout

86. Binary encoding uses:

- a) minimum flip-flops**
- b) maximum flip-flops
- c) one-hot encoding
- d) gray code

87. One-hot encoding uses:

- a) 1 flip-flop
- b) binary logic
- c) one flip-flop per state**
- d) no flip-flops

88. Mealy machine RTL contains:

- a) only combinational logic
- b) combinational + sequential logic**
- c) only sequential logic
- d) no clock

89. State register is implemented using:

- a) gates
- b) multiplexers
- c) flip-flops**
- d) encoders

90. Output glitches occur if:

- a) inputs change near clock edge**
- b) clock is slow
- c) reset is synchronous
- d) power is high

91. FSM verification ensures:

- a) power optimization
- b) correct state transitions**
- c) layout rules
- d) yield improvement

92. Mealy FSM latency is:

- a) lower than Moore FSM**
- b) higher than Moore FSM
- c) same as Moore FSM
- d) infinite

93. FSM synthesis converts:

- a) netlist to RTL
- b) RTL to gate-level logic**
- c) layout to silicon
- d) GDSII to chip

94. Improper FSM design may cause:

- a) faster clock
- b) unintended states**
- c) lower area
- d) better timing

95. Default state in FSM is set using:

- a) clock
- b) reset signal**
- c) input
- d) output

96. UART stands for:

- a) universal asynchronous receiver transmitter**
- b) universal analog receiver transmitter
- c) unified asynchronous register transfer
- d) universal advanced real-time

97. UART is:

- a) asynchronous communication protocol**
- b) synchronous protocol
- c) parallel protocol
- d) optical protocol

98. UART uses how many data lines?

- a) one
- b) two (tx and rx)**
- c) four
- d) eight

99. UART communication does not require:

- a) clock line**
- b) baud rate
- c) tx line
- d) rx line

100. Baud rate refers to:

- a) voltage level
- b) bits per second**
- c) frame size
- d) parity type

101. Start bit value is:

- a) logic 0**
- b) logic 1
- c) high impedance
- d) undefined

102. Stop bit value is:

- a) logic 0
- b) logic 1**
- c) parity bit
- d) clock



103. UART frame contains:

- a) only data bits
- b) start and data only
- c) start, data, optional parity, stop**
- d) data and clock

104. Parity bit is used for:

- a) speed control
- b) error detection**
- c) flow control
- d) synchronization

105. Common UART data length is:

- a) 4 bits
- b) 6 bits
- c) 8 bits**
- d) 16 bits

106. UART full-duplex means:

- a) transmit and receive simultaneously**
- b) transmit only
- c) receive only
- d) half speed

107. UART is widely used in:

- a) memory design
- b) serial communication**
- c) power electronics
- d) RF circuits

108. Baud rate mismatch causes:

- a) faster transfer
- b) data corruption**
- c) lower power
- d) no effect

109. UART idle line is at:

- a) logic 0
- b) logic 1**
- c) floating
- d) undefined

110. UART does not support:

- a) parity
- b) stop bits
- c) clock recovery**
- d) data framing

111. UART transmitter converts:

- a) parallel data to serial data**
- b) serial to parallel
- c) analog to digital
- d) clock to data

112. UART receiver converts:

- a) parallel to serial
- b) serial to parallel**
- c) analog to digital
- d) clock to data

113. UART transmitter block includes:

- a) comparator
- b) shift register**
- c) decoder
- d) latch only

114. UART receiver samples data at:

- a) random time
- b) middle of bit period**
- c) end of frame
- d) start bit only

115. Baud rate generator provides:

- a) reset
- b) timing reference**
- c) parity
- d) clock tree

116. Oversampling improves:

- a) power
- b) noise immunity**
- c) area
- d) clock skew

117. UART transmission is LSB first because:

- a) standard convention**
- b) hardware limitation
- c) power saving
- d) faster speed

118. Frame error occurs if:

- a) stop bit is invalid**
- b) parity fails
- c) data mismatch
- d) baud matches

119. UART FIFO is used to:

- a) increase speed
- b) buffer data**
- c) generate clock
- d) reduce area

120. Flow control prevents:

- a) noise
- b) data loss**
- c) parity error
- d) delay



121. UART tx enable signal controls:

a) data transmission

b) reception

c) reset

d) parity

122. UART rx ready indicates:

a) tx busy

b) data received

c) error occurred

d) reset active

123. UART is commonly used in:

a) DDR memory

b) microcontroller communication

c) GPU design

d) clock synthesis

124. UART design is mostly:

a) FSM-based

b) purely combinational

c) analog

d) asynchronous logic only

125. UART throughput depends on:

a) voltage

b) baud rate

c) layout

d) temperature

126. Verilog is a:

a) hardware description language

b) programming language only

c) scripting language

d) assembly language

127. Verilog describes:

- a) software flow
- b) digital hardware behavior**
- c) analog circuits only
- d) layouts

128. Module is the basic unit in:

- a) verilog**
- b) spice
- c) c language
- d) python

129. Verilog file extension is:

- a) .txt
- b) .hdl
- c) .v**
- d) .svh

130. Input/output ports are declared using:

- a) assign
- b) input, output**
- c) always
- d) wire only

131. wire data type is used for:

- a) combinational connections**
- b) storage
- c) clocking
- d) loops

132. reg data type is used for:

- a) physical wires
- b) storing values**
- c) layout
- d) clocks only

133. always block is used for:

- a) continuous assignment
- b) procedural logic**
- c) module instantiation
- d) simulation control

134. assign statement is:

- a) continuous assignment**
- b) procedural
- c) sequential
- d) clocked

135. Behavioral modeling uses:

- a) gate primitives
- b) always blocks**
- c) layout cells
- d) test vectors

136. Structural modeling uses:

- a) gate/module instantiation**
- b) always block
- c) initial block
- d) loops

137. begin-end is used to:

- a) declare module
- b) group multiple statements**
- c) define ports
- d) generate clock

138. Testbench is used for:

- a) synthesis
- b) verification**
- c) layout
- d) fabrication

139. Testbench contains:

a) input/output ports

b) no ports

c) physical pins

d) clocks only

140. initial block executes:

a) once at simulation start

b) repeatedly

c) at every clock

d) after reset

141. \$monitor is used to:

a) stop simulation

b) display signal values continuously

c) generate waveform

d) reset design

142. \$finish command:

a) pauses simulation

b) ends simulation

c) restarts simulation

d) displays output

143. Testbench clock is generated using:

a) assign

b) always block with delay

c) input port

d) module

144. Delay operator in Verilog is:

a) @

b) #

c) \$

d) %

145. Sensitivity list is used in:

- a) assign
- b) always block**
- c) module
- d) wire

146. Non-blocking assignment operator is:

- a) =
- b) <=**
- c) ==
- d) :=

147. Blocking assignment operator is:

- a) =**
- b) <=
- c) ==
- d) !=

148. Simulation checks:

- a) layout rules
- b) functional correctness**
- c) power only
- d) yield

149. Verilog testbench is:

- a) synthesizable
- b) non-synthesizable**
- c) hardware
- d) layout

150. Verilog helps designers to:

- a) fabricate chips
- b) model and verify digital designs**
- c) draw layouts
- d) test wafers

151. The main purpose of a testbench is to:

- a) synthesize hardware
- b) verify design functionality**
- c) reduce power
- d) generate layout

152. A testbench is usually:

- a) synthesizable
- b) non-synthesizable**
- c) hardware dependent
- d) gate-level

153. Stimulus in testbench refers to:

- a) outputs
- b) input test vectors**
- c) clock tree
- d) layout rules

154. Testbench typically contains:

- a) input and output ports
- b) no ports**
- c) only outputs
- d) only inputs

155. Clock generation in testbench is done using:

- a) assign
- b) always block**
- c) module instantiation
- d) wire

156. Reset is applied in testbench using:

- a) assign
- b) initial block**
- c) module
- d) sensitivity list

157. Delays in stimulus are given using:

- a) @
- b) # operator**
- c) <=
- d) ==

158. Testbench drives signals declared as:

- a) wire
- b) reg**
- c) integer only
- d) parameter

159. Self-checking testbench includes:

- a) manual checking
- b) assertions or comparisons**
- c) only waveforms
- d) clock only

160. \$display is used to:

- a) end simulation
- b) print values once**
- c) loop execution
- d) drive signals

161. \$monitor differs from \$display because it:

- a) prints continuously on change**
- b) prints once
- c) ends simulation
- d) delays execution

162. Functional coverage checks:

- a) timing violations
- b) exercised functionality**
- c) layout errors
- d) power consumption

163. A good testbench should be:

- a) slow
- b) complex
- c) repeatable and automated**
- d) manual

164. Testbench stimulus should cover:

- a) only normal cases
- b) normal and corner cases**
- c) outputs only
- d) reset only

165. Simulation ends using:

- a) \$stop
- b) \$finish**
- c) \$monitor
- d) \$display

166. APB stands for:

- a) advanced peripheral bus**
- b) advanced pipeline bus
- c) asynchronous peripheral bus
- d) advanced processor bus

167. APB is part of:

- a) PCI
- b) AMBA architecture**
- c) AXI only
- d) UART

168. APB is mainly used for:

- a) high-speed data
- b) low-power peripherals**
- c) graphics
- d) memory access



169. APB is a:

- a) pipelined bus
- b) non-pipelined bus**
- c) serial bus
- d) asynchronous bus

170. APB supports:

- a) burst transfers
- b) single-cycle transfers**
- c) cache coherency
- d) out-of-order execution

171. APB interface is:

- a) simple and low complexity**
- b) high throughput
- c) multi-master
- d) clockless

172. APB operates using:

- a) dual clock
- b) single clock**
- c) no clock
- d) recovered clock

173. APB is suitable for:

- a) DRAM
- b) UART, GPIO, timers**
- c) CPU core
- d) GPU

174. APB protocol is optimized for:

- a) performance
- b) low power and simplicity**
- c) latency hiding
- d) bandwidth



175. APB transfer consists of:

- a) one phase
- b) setup and access phases**
- c) three phases
- d) pipeline stages

176. APB does not support:

- a) read
- b) write
- c) burst transactions**
- d) error signaling

177. APB slaves are selected using:

- a) address decoder
- b) PSEL signal**
- c) PENABLE
- d) PREADY

178. APB protocol is defined by:

- a) IEEE
- b) ARM**
- c) ISO
- d) JEDEC

179. APB timing is:

- a) synchronous**
- b) asynchronous
- c) source synchronous
- d) self-timed

180. APB simplifies peripheral interfacing by:

- a) increasing speed
- b) reducing control signals**
- c) adding pipeline
- d) adding arbitration

181. PCLK in APB is used for:

- a) reset
- b) clocking transfers**
- c) selecting slave
- d) error handling

182. PRESETn is:

- a) active-low reset**
- b) active-high reset
- c) clock enable
- d) interrupt

183. PADDR carries:

- a) data
- b) address information**
- c) control signals
- d) parity

184. PWRITE indicates:

- a) read operation
- b) write operation when high**
- c) slave select
- d) reset

185. PWDATA carries:

- a) address
- b) write data**
- c) read data
- d) error code

186. PRDATA carries:

- a) write data
- b) read data**
- c) address
- d) reset



187. PENABLE is asserted during:

- a) idle phase
- b) access phase**
- c) reset
- d) decode

188. PREADY indicates:

- a) slave selected
- b) transfer completion**
- c) error
- d) reset

189. PSLVERR is used for:

- a) data transfer
- b) reset
- c) error indication**
- d) clocking

190. APB setup phase occurs when:

- a) PSEL is high and PENABLE is low**
- b) both high
- c) both low
- d) PREADY high

191. APB access phase occurs when:

- a) PSEL low
- b) PSEL and PENABLE high**
- c) only PENABLE high
- d) reset active

192. Address and control signals are sampled during:

- a) access phase
- b) setup phase**
- c) reset
- d) idle

193. Data transfer completes when:

- a) PSEL goes low
- b) PREADY is high during access**
- c) reset asserted
- d) PENABLE low

194. APB allows wait states using:

- a) PENABLE
- b) PREADY**
- c) PSEL
- d) PSLVERR

195. APB timing is deterministic because it:

- a) is not pipelined**
- b) supports bursts
- c) is asynchronous
- d) uses arbitration

196. APB master initiates:

- a) all bus transactions**
- b) only read
- c) only write
- d) reset

197. APB master FSM controls:

- a) clock
- b) setup and access phases**
- c) slave logic
- d) reset

198. APB master design starts with:

- a) coding
- b) protocol understanding**
- c) synthesis
- d) layout

199. APB master generates:

- a) PRDATA
- b) PADDR, PSEL, PWRITE**
- c) PSLVERR
- d) PREADY

200. APB master must deassert PENABLE after:

- a) setup phase
- b) transfer completion**
- c) reset
- d) idle

201. APB master FSM states include:

- a) fetch, decode
- b) idle, setup, access**
- c) read, write only
- d) reset only

202. APB master Verilog uses:

- a) FSM-based design**
- b) combinational only
- c) analog blocks
- d) latches only

203. PSEL should be asserted:

- a) after PENABLE
- b) before PENABLE**
- c) with reset
- d) at end

204. In Verilog, APB master outputs are declared as:

- a) wire
- b) reg**
- c) integer
- d) parameter

205. APB master read operation sets PWRITE to:

- a) 0
- b) 1
- c) z
- d) x

206. APB master write operation sets PWRITE to:

- a) 0
- b) 1**
- c) z
- d) x

207. FSM transition depends on:

- a) clock only
- b) PREADY and current state**
- c) data only
- d) reset only

208. APB master holds address stable during:

- a) idle
- b) setup and access**
- c) reset
- d) decode

209. Incorrect APB master design may cause:

- a) faster transfer
- b) protocol violation**
- c) low power
- d) reduced area

210. APB master verification checks:

- a) synthesis
- b) timing and protocol compliance**
- c) layout
- d) yield

211. ALU stands for:

- a) **arithmetic logic unit**
- b) advanced logic unit
- c) arithmetic level unit
- d) array logic unit

212. ALU performs:

- a) storage
- b) **arithmetic and logic operations**
- c) memory access
- d) control flow

213. Common ALU operations include:

- a) load/store
- b) **add, subtract, AND, OR**
- c) branching
- d) reset

214. ALU inputs are typically:

- a) one operand
- b) **two operands**
- c) three operands
- d) none

215. ALU output flags include:

- a) reset
- b) **zero, carry, overflow**
- c) address
- d) clock

216. ALU operation is selected using:

- a) clock
- b) **control signal/opcode**
- c) reset
- d) data only

217. ALU is a:

- a) combinational circuit**
- b) sequential circuit
- c) analog block
- d) memory

218. Verilog ALU uses:

- a) case statement**
- b) initial only
- c) always @(*) missing
- d) clocked logic only

219. ALU subtraction is implemented using:

- a) AND gate
- b) adder with 2's complement**
- c) multiplier
- d) divider

220. Overflow occurs when:

- a) operands are zero
- b) result exceeds bit width**
- c) reset active
- d) clock slow

221. ALU carry flag indicates:

- a) negative result
- b) carry out from MSB**
- c) parity
- d) overflow

222. Zero flag is set when:

- a) result is negative
- b) result equals zero**
- c) carry occurs
- d) reset active

223. ALU Verilog module ports include:

- a) only inputs
- b) inputs and outputs**
- c) no ports
- d) clock only

224. ALU design verification requires:

- a) synthesis
- b) exhaustive test vectors**
- c) layout
- d) fabrication

225. ALU speed depends on:

- a) layout only
- b) adder architecture**
- c) reset
- d) parity

226. Half-adder adds:

- a) two single-bit inputs**
- b) three inputs
- c) multi-bit numbers
- d) signed numbers

227. Half-adder outputs are:

- a) sum and borrow
- b) sum and carry**
- c) difference and carry
- d) quotient and remainder

228. Sum output of half-adder is:

- a) XOR of inputs**
- b) AND of inputs
- c) OR of inputs
- d) XNOR



229. Carry output of half-adder is:

- a) XOR
- b) AND**
- c) OR
- d) NOR

230. Half-adder is a:

- a) combinational circuit**
- b) sequential circuit
- c) memory
- d) controller

231. I2C stands for:

- a) inter-integrated circuit**
- b) integrated interface control
- c) internal interrupt controller
- d) interconnect interface circuit

232. SDA in I2C carries:

- a) clock
- b) serial data**
- c) address only
- d) power

233. SCL in I2C carries:

- a) serial clock**
- b) data
- c) reset
- d) interrupt

234. I2C bus is:

- a) point-to-point
- b) multi-master, multi-slave**
- c) simplex
- d) parallel



235. I2C start condition occurs when:

- a) SDA goes low while SCL is high**
- b) SDA high while SCL low
- c) both low
- d) both high

236. I2C stop condition occurs when:

- a) SDA low, SCL high
- b) SDA goes high while SCL is high**
- c) both low
- d) reset active

237. SDA line is:

- a) bidirectional**
- b) input only
- c) output only
- d) clock

238. I2C uses which type of outputs?

- a) push-pull
- b) open-drain**
- c) tri-state only
- d) analog

239. Histogram framer is used to:

- a) compress data
- b) count frequency of values**
- c) generate clock
- d) reset system

240. Histogram bins represent:

- a) memory addresses
- b) value ranges**
- c) clock cycles
- d) instructions

241. Histogram design uses:

- a) latches only
- b) counters and comparators**
- c) analog blocks
- d) FSM only

242. Histogram output shows:

- a) timing
- b) data distribution**
- c) power
- d) layout

243. Final design review ensures:

- a) faster coding
- b) functional correctness and completeness**
- c) higher voltage
- d) larger area

244. Testing verifies design against:

- a) layout
- b) specifications**
- c) fabrication
- d) power grid

245. Mini demonstration shows:

- a) RTL only
- b) working functionality**
- c) layout
- d) synthesis report

246. Waveform analysis helps in:

- a) fabrication
- b) debugging behavior**
- c) power routing
- d) floorplanning

247. Bug fixing is easiest at:

- a) fabrication
- b) layout
- c) simulation stage**
- d) packaging

248. Integration testing checks:

- a) individual module only
- b) interaction between modules**
- c) layout rules
- d) clock tree

249. Documentation is important for:

- a) simulation speed
- b) design clarity and maintenance**
- c) power reduction
- d) clock generation

250. Final validated design is:

- a) directly fabricated
- b) ready for further implementation stages**
- c) discarded
- d) re-coded

Let's Learn and Grow